

FRAUD DETECTION AND RISK ANALYSIS USING SQL

Tools & Technologies

PostgreSQL | pgAdmin 4 | SQL | Microsoft Word

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Project Overview:

Financial fraud is a growing challenge for banks and financial institutions, leading to significant financial losses and security concerns. Detecting fraudulent transactions and understanding the factors associated with fraud are essential for reducing risk and improving customer security.

The objective of this project is to analyze financial transaction data and customer account information to identify fraud patterns, evaluate risk factors, and generate actionable business insights. Using SQL and PostgreSQL, the project examines transaction behavior, customer profiles, security measures, and fraud indicators to understand the characteristics associated with fraudulent activities.

The analysis is based on two datasets: a transaction dataset containing transaction-level information and an account profile dataset containing customer account details. Data cleaning, exploratory data analysis, and join-based analysis were performed to investigate the relationship between customer characteristics and fraud occurrence.

Objectives:

- Analyze fraudulent and non-fraudulent transactions.
- Identify high-risk customer and transaction characteristics.
- Evaluate the impact of security measures such as Two-Factor Authentication (2FA).
- Examine fraud patterns across merchants, countries, and account types.
- Generate recommendations to help reduce fraud risk.

Tools Used:

- PostgreSQL
- SQL
- pgAdmin 4

Dataset Information:

The project utilizes two datasets:

Account Profiles Dataset:

Customer account information

Account age

Credit limits

Risk scores

Account type

Security attributes (2FA)

Transactions Dataset:

Transaction details

Merchant information

Transaction amounts

Device information

Risk indicators

Fraud labels

Dataset Overview:

This project uses two related datasets: Account Profiles and Transactions. The datasets are linked through the "account_id" field, allowing transaction behavior to be analyzed alongside customer account characteristics.

1. Account Profiles Dataset

The Account Profiles dataset contains customer account information and risk-related attributes. It provides details about account characteristics, security measures, transaction behavior summaries, and fraud indicators.

Key Variables:

Column Name	Description
account_id	Unique identifier for each customer account
account_age_days	Age of the account in days
credit_limit	Maximum credit limit assigned to the account
home_country	Customer's registered country
risk_score	Risk score assigned to the account
is_high_risk	Indicates whether the account is classified as high risk
avg_txn_amount	Average transaction amount
avg_monthly_txns	Average monthly transactions
has_2fa	Indicates whether Two-Factor Authentication is enabled
account_type	Type of account
total_transactions	Total number of transactions
fraud_count	Number of fraudulent transactions associated with the account
fraud_rate	Percentage of fraudulent transactions associated with the account

Purpose:

This dataset helps identify customer-level risk factors and evaluate how account characteristics influence fraud occurrence.

2. Transactions Dataset:

The Transactions dataset contains individual transaction records and transaction-specific risk indicators. It provides information about transaction behavior, merchant activity, device usage, and fraud labels.

Key Variables:

Column name	Description
transaction_id	Unique identifier for each transaction
account_id	Customer account associated with the transaction
transaction_timestamp	Date and time of transaction
amount	Transaction amount
merchant	Merchant involved in the transaction
merchant_country	Country where the merchant is located
device_type	Device used for the transaction
card_present	Indicates whether the card was physically present
ip_risk_score	Transaction risk score
is_foreign	Indicates whether the transaction was foreign
velocity_1h	Number of recent transactions within one hour
amount_vs_avg	Transaction amount compared to customer average
is_fraud	Fraud indicator (1 = Fraud, 0 = Non-Fraud)

fraud_pattern | Fraud category or pattern label

Purpose:

This dataset helps identify transaction-level fraud patterns, suspicious activities, and behavioral indicators associated with fraudulent transactions.

Dataset Relationship

The two datasets are connected using the "account_id" column.

By joining these datasets, it becomes possible to analyze how customer characteristics such as account type, account age, credit limits, and security measures influence fraudulent transaction behavior.

This relationship forms the foundation for the fraud detection and risk analysis performed in this project.

Data Cleaning and Validation:

Before performing the analysis, both datasets were examined to ensure data quality and consistency.

The following checks were performed:

Account Profiles Dataset

- Checked for duplicate account IDs.
- Checked for missing values in key columns.
- Validated risk score values.
- Validated credit limit values.
- Verified account age values.
- Verified 2FA indicator values.

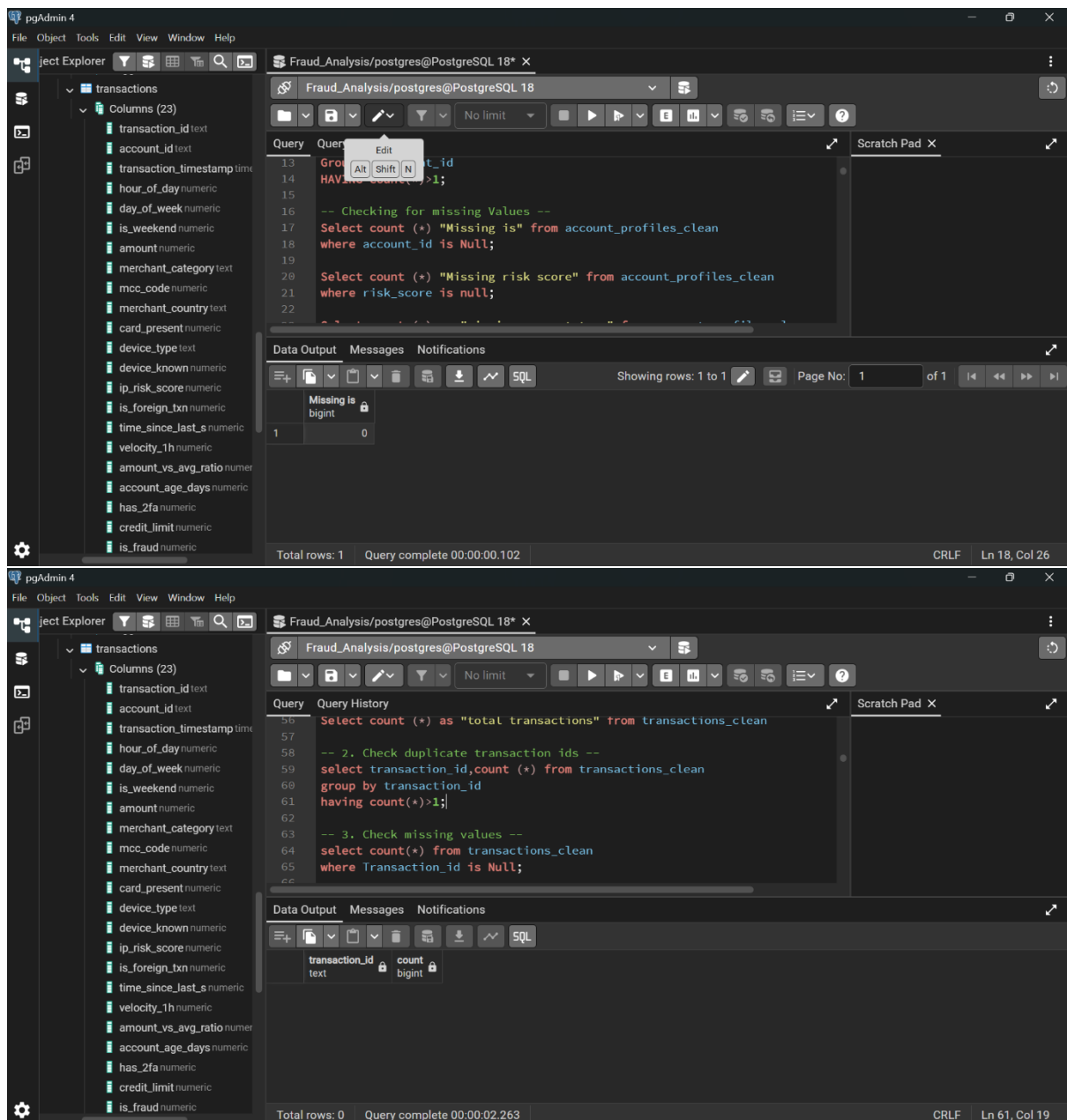
Transactions Dataset

- Checked for duplicate transaction IDs.
- Checked for missing values in key columns.
- Validated transaction amounts.
- Verified fraud indicator values.
- Validated risk-related variables.
- Confirmed successful mapping between transactions and account profiles using account_id.

Results

- No duplicate records were identified.
- No significant missing values were detected in critical fields.
- Numeric fields were within expected ranges.
- Fraud indicators contained valid values.
- The datasets were successfully joined using account_id.

The datasets were considered clean and suitable for further analysis.



Exploratory Data Analysis (EDA) Summary:

The exploratory data analysis was conducted to understand the structure, characteristics, and distribution of data within both the Account Profiles and Transactions datasets before performing fraud analysis.

Account Profiles Dataset:

The Account Profiles dataset contains customer account information, risk indicators, and security-related attributes.

Key Observations:

- The dataset contained 50000 customer accounts.
- The most common account type was Personal, while premium represented the smallest account segment.

- Approximately 64.87 % of accounts had Two-Factor Authentication (2FA) enabled.
- The average account risk score was 19.912278.
- A total of 18 accounts were classified as high-risk.
- The average credit limit across all accounts was 6724.5271190.
- The average account age was 1837.4661 days.

These observations provided valuable insights into customer demographics, account security adoption, and overall risk distribution.

Transactions Dataset:

The Transactions dataset contains transaction-level information, merchant details, transaction risk indicators, and fraud labels.

Key Observations:

- The dataset contained 1000000 transactions.
- A total of 17,143 transactions were identified as fraudulent.
- The overall fraud rate was 1.71 %.
- US country is recorded higher fraud activity compared to other countries.
- Fraud occurrence varied across different device types.
- Foreign transactions showed different fraud patterns compared to domestic transactions.
- Merchants exhibited varying levels of fraud exposure, with some merchants accounting for a larger share of fraudulent activity.

These findings provided an initial understanding of transaction behavior and highlighted potential areas requiring deeper fraud analysis.

Summary:

The exploratory analysis revealed meaningful variations in customer risk profiles, account security measures, transaction behavior, and fraud distribution. These findings established the foundation for the subsequent fraud analysis and join-based investigation between customer accounts and transactions.

Fraud Risk Analysis:

1. Does 2FA reduce fraud?

SQL query:

```
select a.has_2fa,
count (*) as "transactions",
sum(t.is_fraud) as "Fraud_cases",
round(100*sum(t.is_fraud)/count(*),2) as "fraud rate"
from transactions_clean as "t"
join account_profiles_clean as "a"
on t.account_id=a.account_id
group by a.has_2fa;
```

Output:

Data Output		Messages	Notifications	
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	has_2fa numeric	transactions bigint	Fraud_cases numeric	fraud rate numeric
1	0	352433	7160	2.03
2	1	647567	9983	1.54

Insight:

Accounts without Two-Factor Authentication (2FA) experienced a higher fraud rate (2.03%) compared to accounts with 2FA enabled (1.54%). Although accounts with 2FA generated a larger number of transactions, their fraud rate remained lower. This suggests that enabling 2FA can help reduce the likelihood of fraudulent activity and improve account security.

2. Which account type has more frauds?

SQL Query:

```
select a.account_type,  
       count (*) as "transactions",  
       sum(t.is_fraud) as "Fraud_cases", round(100*sum(t.is_fraud)/count(*),2) as "fraud rate"  
from transactions_clean as "t"  
join account_profiles_clean as "a"  
on t.account_id=a.account_id  
group by a.account_type;
```

Output:

Data Output		Messages	Notifications	
</				

Insight:

Business accounts recorded the highest fraud rate at 1.76%, followed by Personal accounts at 1.71%, while Premium accounts had the lowest fraud rate at 1.63%. Although Personal accounts accounted for the largest number of fraud cases (12,062) due to their higher transaction volume, Business accounts were relatively more vulnerable to fraudulent activity when considering fraud rate. This suggests that business accounts may require stronger fraud monitoring and security controls.

3. Do high risk account actually experience more fraud?

SQL Query:

```
select a.is_high_risk,  
       count (*) as "transactions",
```

```

sum(t.is_fraud) as "Fraud_cases", round(100*sum(t.is_fraud)/count(*),2) as "fraud rate"
from transactions_clean as "t"
join account_profiles_clean as "a"
on t.account_id=a.account_id
group by a.is_high_risk;

```

Output:

	is_high_risk numeric	transactions bigint	Fraud_cases numeric	fraud rate numeric
1	0	999674	17126	1.71
2	1	326	17	5.21

Insight:

Accounts classified as high-risk recorded a fraud rate of 5.21%, compared to only 1.71% for non-high-risk accounts. Although high-risk accounts represented a very small portion of total transactions (326 transactions), they were approximately 3 times more likely to experience fraud. This indicates that the risk scoring system is effective in identifying accounts with elevated fraud exposure and can be used as an important indicator for fraud monitoring and prevention.

4. Does Account age matter?

Select case

```

WHEN a.account_age_days <= 365 then 'New'
when a.account_age_days > 365 and a.account_age_days < 1825 then 'Medium'
else 'old'
end as "Account",
count(*) as "transaction",
sum(t.is_fraud) as "Fraud_cases"
from transactions_clean as "t"
join account_profiles_clean as "a"
on t.account_id=a.account_id
group by
case
WHEN a.account_age_days <= 365 then 'New'
when a.account_age_days > 365 and a.account_age_days < 1825 then 'Medium'
else 'old'
end ;

```


Output:

	Account	transaction	Fraud_cases
	text	bigint	numeric
1	Medium	404760	6824
2	New	93378	1785
3	old	501862	8534

Insight:

Older accounts generated the highest number of transactions (501,862) and also recorded the highest number of fraud cases (8,534), followed by medium-aged accounts. New accounts had the lowest transaction volume (93,378) and the fewest fraud cases (1,785). This suggests that fraud occurrences increase with transaction activity and account maturity. However, since older accounts also process substantially more transactions, fraud rates should be analyzed alongside transaction volume to determine true risk exposure.

5. Risk Score

SQL Query:

```
select round(avg(a.risk_score),2) as "avg_account_risk",
round(avg(t.ip_risk_score),2) as "avg_transaction_risk"
from transactions_clean as "t"
join account_profiles_clean as "a"
on t.account_id=a.account_id
where t.is_fraud =1;
```

Output:

	avg_account_risk	avg_transaction_risk
	numeric	numeric
1	19.95	55.35

Insight:

Fraudulent transactions had an average account risk score of 19.95 and an average transaction risk score of 55.35. The significantly higher transaction risk score suggests that suspicious transaction behavior is more closely associated with fraud than the overall risk profile of the account.

Key Findings

- Accounts that used 2FA experienced fewer fraud incidents compared to accounts without 2FA. This shows that an extra layer of authentication can help reduce fraud risk.

- Among the different account types, business accounts had the highest fraud rate, making them more vulnerable to fraudulent activities than personal or premium accounts.
- Accounts marked as high-risk showed a much higher fraud rate than normal accounts. This indicates that the risk classification system is effective in identifying potentially risky accounts.
- Older accounts recorded the highest number of transactions and fraud cases. However, this is likely because older accounts are more active and process more transactions overall.
- Fraudulent transactions had a significantly higher transaction risk score compared to the average account risk score. This suggests that transaction behaviour is a stronger indicator of fraud than account characteristics alone.
- Fraud cases were found across all account segments, but they represented only a small portion of the total transactions in the dataset.

Recommendations

- Encourage users to enable Two-Factor Authentication (2FA) to improve account security.
- Monitor high-risk accounts more closely, as they are more likely to be associated with fraudulent activity.
- Apply additional security checks for business accounts since they showed the highest fraud rate in the analysis.
- Use transaction risk scores as an important factor in fraud detection because they were strongly linked to fraudulent transactions.
- Implement real-time alerts for transactions that appear unusual or have high-risk scores.
- Regularly review and update fraud detection rules based on changing transaction patterns.

Conclusion:

The objective of this project was to analyze account and transaction data to understand the factors associated with fraudulent activity. Using PostgreSQL, the datasets were cleaned, explored, and joined to perform various analyses.

The results showed that fraud is influenced by several factors, including account type, account risk level, authentication methods, and transaction risk scores. Accounts using 2FA were generally safer, while high-risk and business accounts experienced higher levels of fraud. The analysis also revealed that transaction-related risk indicators play an important role in identifying suspicious activity.

Overall, this project demonstrates how SQL can be used to extract meaningful insights from financial data and support fraud detection efforts. The findings can help organizations improve security measures, identify high-risk behaviour, and make better data-driven decisions.

References:

- Kaggle Fraud Detection dataset
- PostgreSQL Documentation
- SQL Learning Resources